

Apr 2026

PROJECT PROPOSAL

Driving Gaming Innovations & Experimentations

Presented by Dell Team X

What is the problem?

Organizations running experiments in complex, high-dimensional environments cannot turn test results into the next best decision fast enough.

Research evidence: why current experimentation breaks down

Literature and industry evidence point to the same bottleneck: experimentation does not scale well when decisions are high-dimensional, fast-moving, and still heavily human-mediated.

01 The manual bottleneck is still real

A 2024 systematic review consolidated 143 studies and found that, despite automation trends, A/B testing still requires significant human involvement.

Quin et al., 2024

02 Scale makes experimentation methodologically harder

A review of online controlled experiments reports that more than 50% of ideas fail to generate meaningful improvements; in large organizations, users may sit in hundreds of concurrent experiments while products branch into billions of versions.

Larsen et al., 2024

03 High-dimensional design spaces exceed testing budgets

Recent work on large-scale factorial experiments shows that digital design spaces are combinatorial while traffic is limited; even a simple $10 \times 5 \times 6 \times 4$ setup yields 1,200 combinations, making brute-force A/B testing operationally difficult.

Wen et al., 2026

04 Mobile gaming shows the business cost clearly

SEGA HARDlight reports that experiment analysis became the limiting factor; once analysis was automated, the team achieved a targeted 2× increase in monthly A/B testing capacity without increasing headcount.

Databricks / HARDlight, 2026

What this means for our capstone

- Firms can collect experiment data and increasingly automate reporting.
- But they still lack a practical way to turn many results into the next best action across interacting variables.
- Our project targets that middle layer: generate, evaluate, and recommend next steps without Tencent-scale infrastructure.

Core problem:
experimentation systems measure outcomes, but they do not close the loop from insight to action in high-dimensional environments.

Adaptive Decision Systems for High-Dimensional Optimization

What triggered this idea (GDC 26' event)

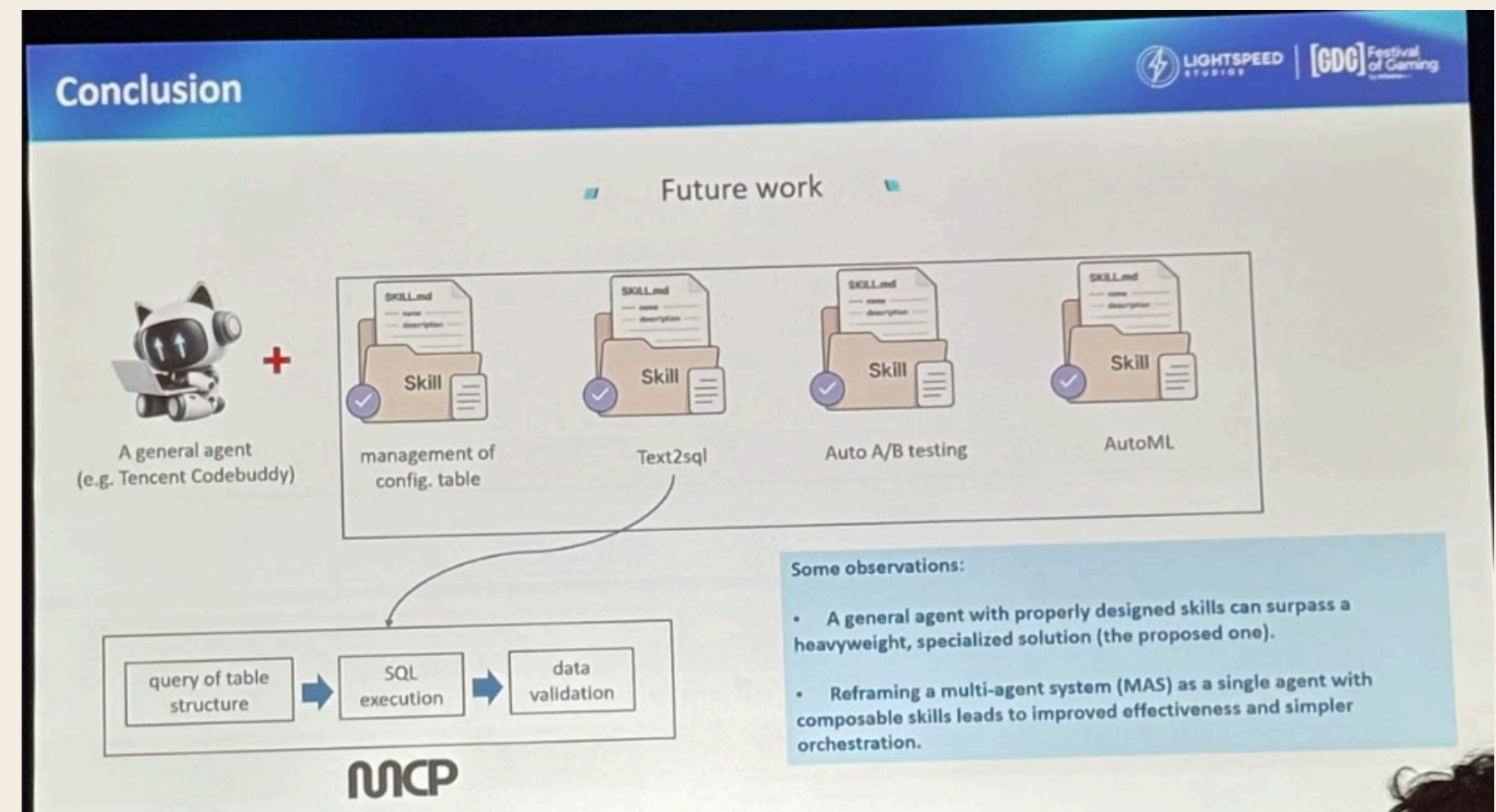
- GDC discussions with Tencent / Lightspeed (Dong Fang)
- Exposure to production systems combining RL + experimentation

Key observation

- Decisions rely on high-dimensional parameter spaces
- A/B testing is too slow and local

Agent's Purpose

Shift from static, manually designed A/B testing toward adaptive systems that continuously explore, evaluate, and optimize decisions in high-dimensional environments (such as video games).



[Dong Fang's presentation slides](#)

Towards Autonomous Optimization Systems

Core mechanism

- Generate decisions (LLM / policies)
- Evaluate impact (experimentation / causal signals)
- Update dynamically (RL / bandits / SFT)
- Closed-loop learning system

Why it matters

- Faster experimental cycles
- Improved global policy optimization
- Proven usage in limited SOTA systems (lightspeed games)
- Transferable to product, marketing, and operational systems

Policy generation → Causal evaluation → Policy update → Continuous learning

Evidence:

<https://arxiv.org/html/2512.02358v1>

<https://www.databricks.com/blog/building-ab-testing-analysis-framework-mobile-gaming-databricks>

Limits of Current Experimentation & Decision Systems

How it works today

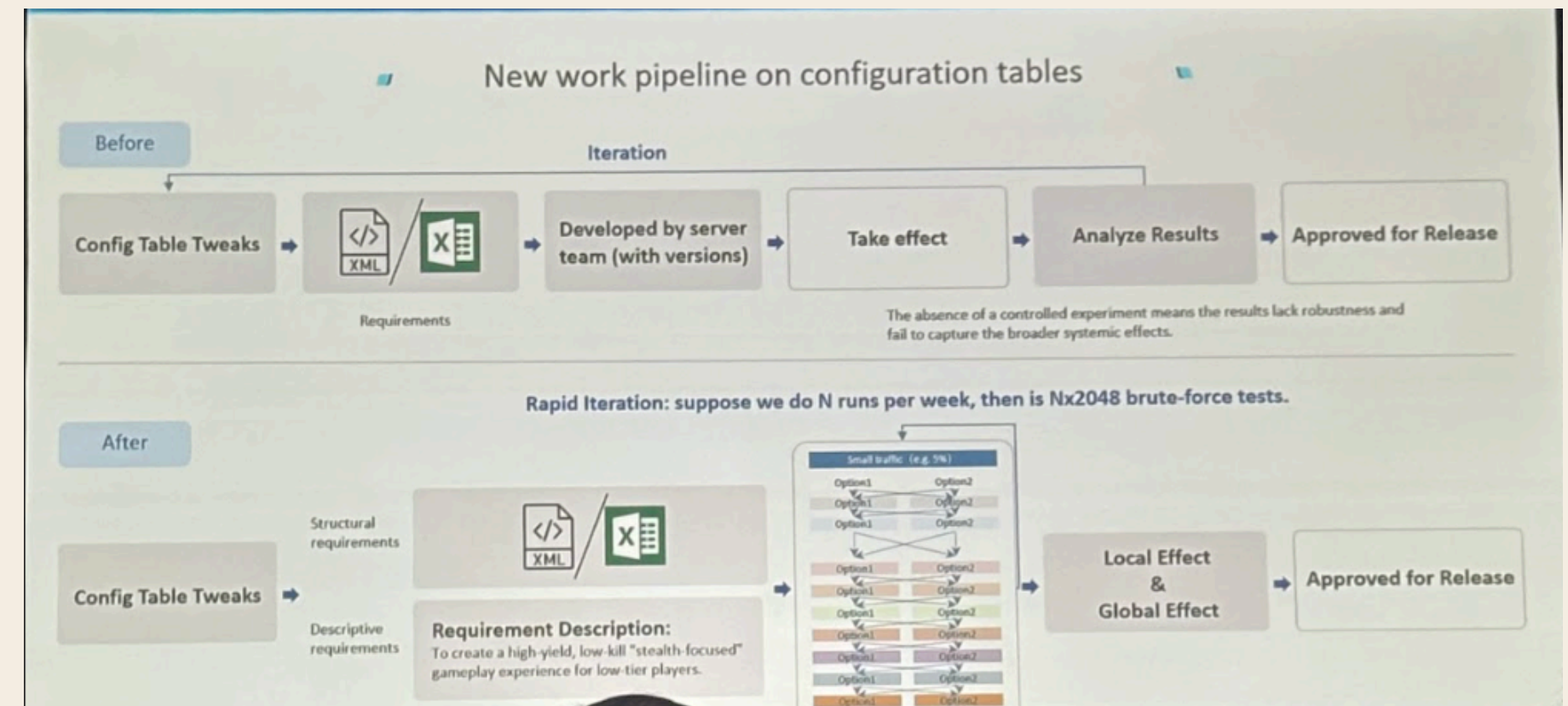
- Manual parameter tuning (configs, heuristics)
- Sequential A/B testing
- Limited exploration of decision space

Consequence:

Systems fail to learn optimal policies in complex, dynamic environments.

Core limitations

- Local optimization only, cannot capture global interactions
- Combinatorial explosion, too many parameter combinations to test
- Slow feedback loops, iteration cycles are costly
- Static decision logic, no continuous adaptation



Use Case from Lightspeed's system on video games configuration tables

The Cost of Slow Experimentation

Why the experimentation bottleneck is a billion-dollar problem

\$178B

**Global gaming industry size,
2024**

Source: Newzoo Global Games Market Report

0.85%

**Median Day 28 retention
in mobile games**

Source: GameAnalytics benchmarks, 10,000+ games

< 0.00003%

**Of Booking.com's config space
covered by 25,000 annual tests**

Source: Booking.com experimentation program

The bottom line

A \$25B user-acquisition spend funnels into games that lose 99% of players in 4 weeks.

Even a 5% retention lift can drive 25%+ profit gains (Bain).

Opportunity Gap

Adaptative Experimentation - Split in Two

Smarter Reporting

- Automates ingestion, stats, and LLM summaries on dashboards.
- Still human-in-the-loop
 - Humans design, read, and decide.
- Closed loop is on their roadmap.

Close-Loop

- Fully generate, evaluate, and update the loop in production.
- Heavyweight and specialized
 - Custom multi-agent system.
- Tencent-scale engineering.

The Gap

An agent that closes the loop without the Tencent-scale overhead

LLM hypothesis generation + causal evaluation + bandit policy updates, built as composable skills on a standard experimentation infrastructure.

Use Case Selection Framework for Agentic Systems

Selecting High-Impact Opportunities for Adaptive Experimentation

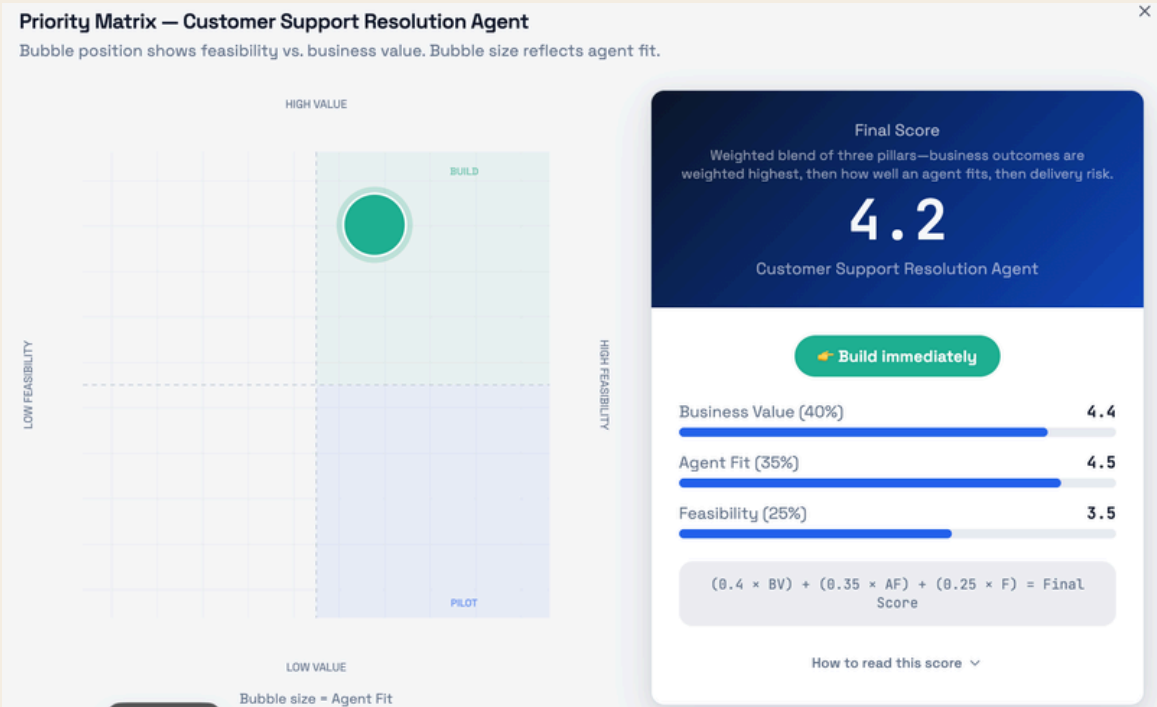
We developed a structured evaluation framework to systematically identify high-impact use cases for agentic systems.

Pillar	What it measures	Why it matters
Agent Fit (35%)	Decision complexity, autonomy, workflow structure	Determines if the problem is suitable for agentic automation
Business Value (40%)	Revenue, cost, time impact, frequency	Quantifies potential impact of solving the problem
Feasibility (25%)	Data readiness, integration effort, risk, speed	Assesses implementation viability

Each use case is scored across sub-metrics within each pillar, and a weighted composite score is calculated to prioritize opportunities

Scoring Framework Application

Final Score = (0.40 × Business Value) + (0.35 × Agent Fit) + (0.25 × Feasibility)



[Link to Scorecard Metrics](#)

Selected Use Case: Adaptive Marketing Experimentation

Applying this framework, adaptive marketing experimentation emerges as a high-priority use case due to:

- High decision frequency and iterative workflows
- Large combinatorial decision space (channels, creatives, targeting)
- Strong data availability from existing experimentation systems
- Clear potential to improve decision speed and quality

System Design: Adaptive Experimentation Agent

How the system generates, evaluates, and updates decisions

Input Data

Structured Data

- Experiment logs (A/B tests, configs)
- Player metrics (retention, engagement, spend)
- Game parameters (difficulty, rewards, UI configs)

Behavioral Data

- Player interaction events (clicks, sessions, progression)

Visual Data

- Gameplay recordings/screen captures
- UI layouts, level design elements
- Player navigation patterns (via visual traces)

Agentic Decision System

Layer 1: Perception Layer (CV + Data Processing)

- Extract gameplay features
- Detect UI friction & difficulty
- Convert visuals → structured signals

Layer 2: Generation (LLM)

- Generates new experiment ideas (level tweaks, UI changes, rewards)
- Uses both structured + visual insights
- Suggests high-potential parameter combinations

Layer 3: Evaluation (Causal)

- Estimates the impact of changes (retention, engagement)
- Filters noise using causal inference/uplift modeling
- Prioritizes statistically meaningful improvements

Layer 4: Optimization (RL / Bandits)

- Updates strategies dynamically
- Learns optimal configurations over time
- Balances exploration vs exploitation

Output / Decisions

- Optimized game configurations (levels, rewards, UI)

- Next-best experiments for player engagement

- Predicted impact on retention / monetization

- Continuous updates to decision policies